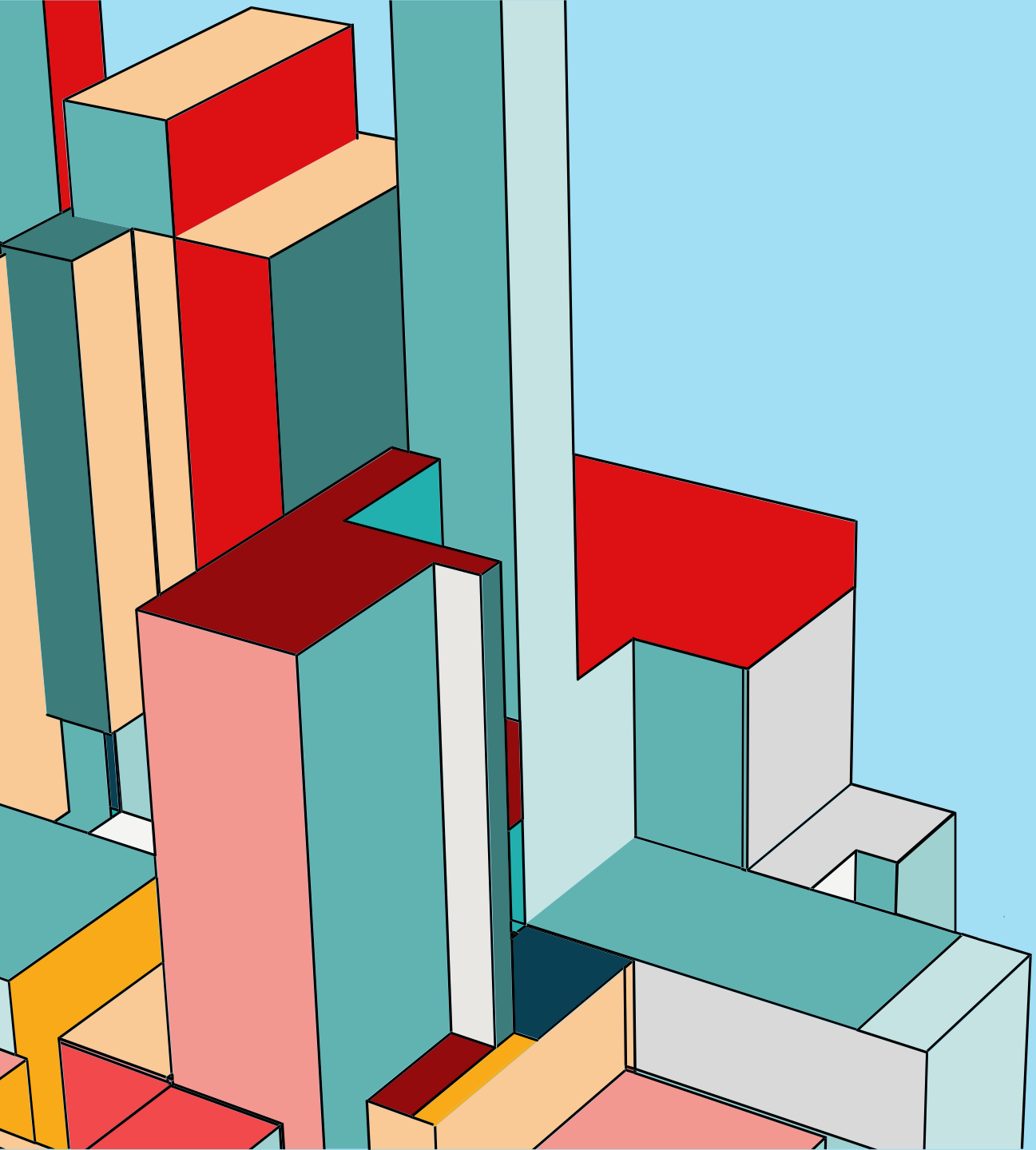


APPLIED DATA SCIENCE PROJECT

FOUNDATION MODELS FOR HIGH-RESOLUTION AIR QUALITY MAPS

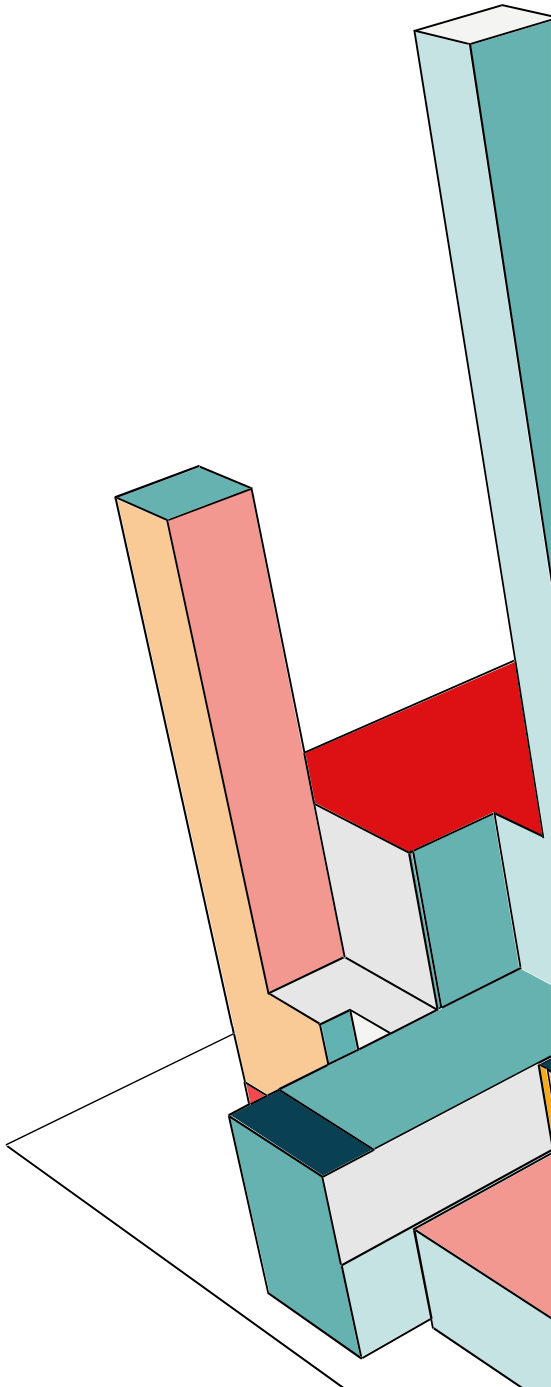
Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

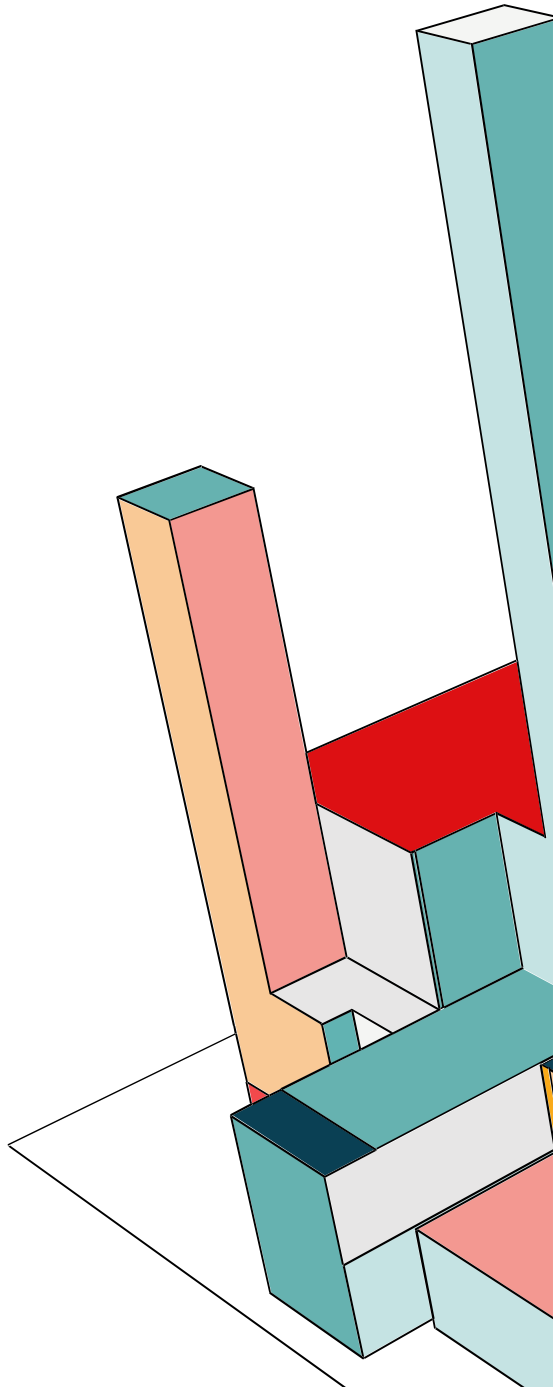
VALUE-DRIVEN PROJECT

- **Air quality** (NO_2, PM2.5, PM10, O_3) has a direct impact on public health, especially in cities, where pollution varies at neighbourhood scale
- **Earth system and EO foundation models** can be fine-tuned for many downstream tasks, but often produce output maps at coarse resolution (e.g. 0.4°, ≈40 km)
- The aim of this project is to benchmark foundation models for **pollutant concentration regression** and to modify one of them to obtain **higher-resolution output maps**



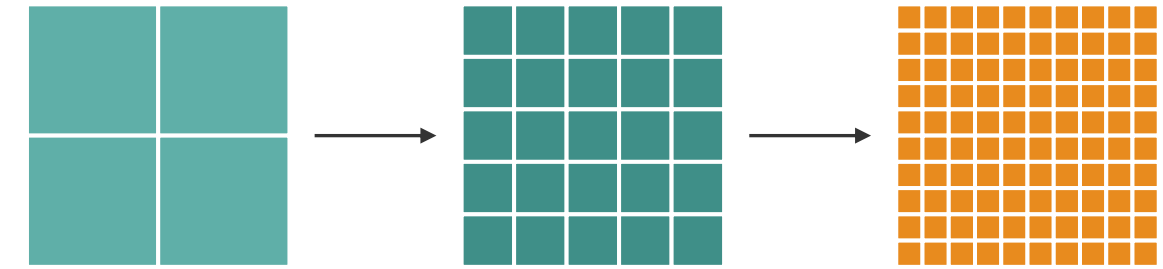
DATA

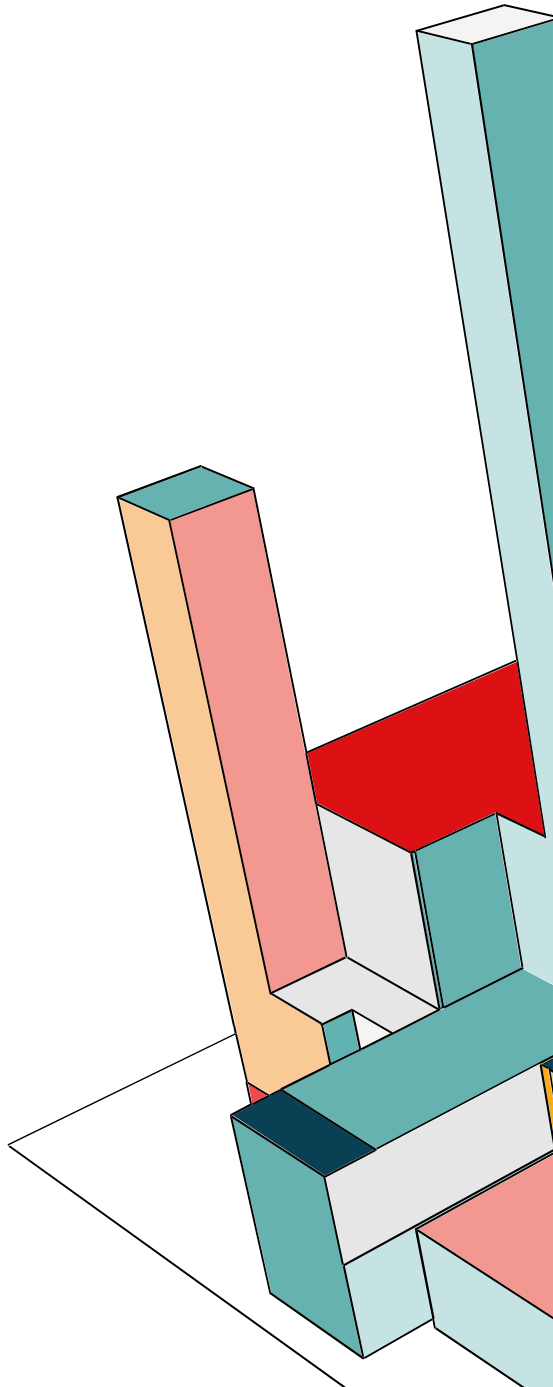
- We will use an **air quality dataset** for pollutant concentration estimation
- The following data are available
    - *Ground-level pollutant concentrations from monitoring stations (Ground truth)*
    - *CAMS atmospheric composition reanalysis and forecasts*
    - *Sentinel-5P TROPOMI images (NO_2, O_3, CO, SO_2)*
    - *Copernicus ERA5 weather data*
    - *Land cover and Copernicus Digital Elevation Model*



TASK

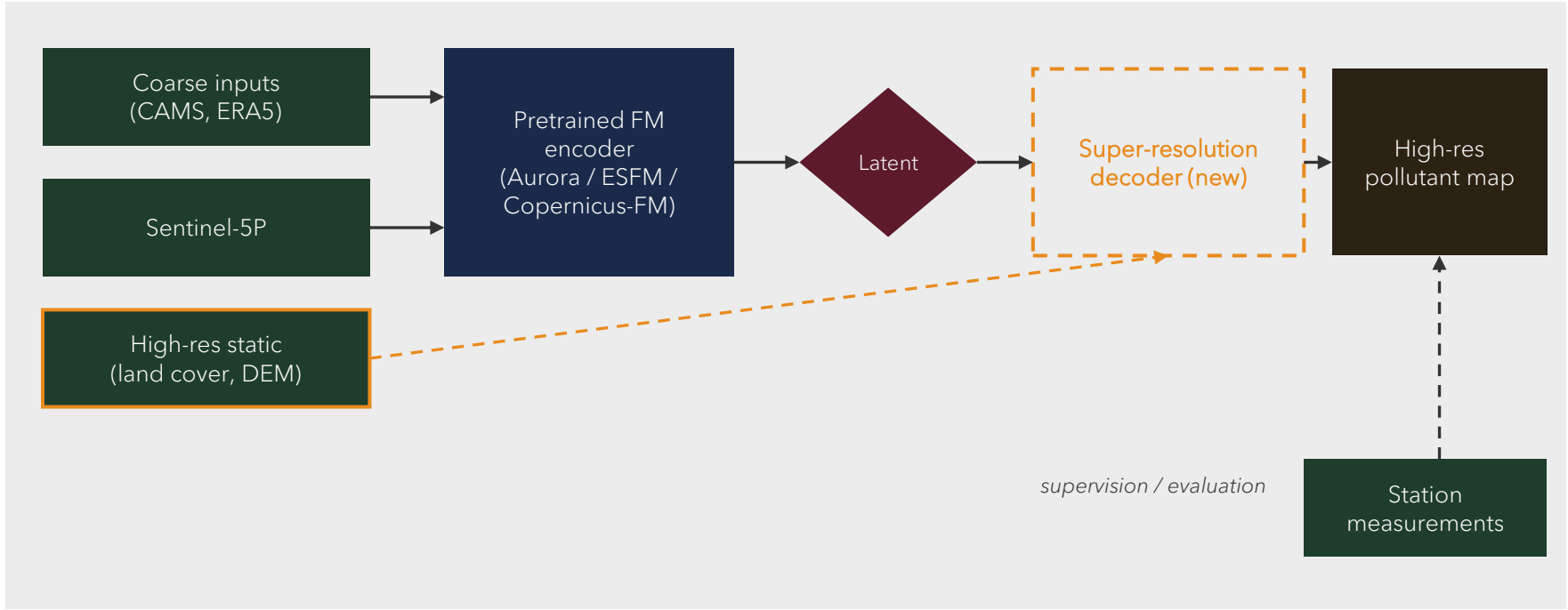
- Assess three foundation models on **pollutant regression**
    - **Aurora**: Earth system FM, air quality at 0.4°
    - **ESFM**: open FM on Aurora's backbone, handles station data
    - **Copernicus-FM**: EO FM for Sentinel-1 to Sentinel-5P
- Compare RMSE, MAE, R^2 across **output resolutions**

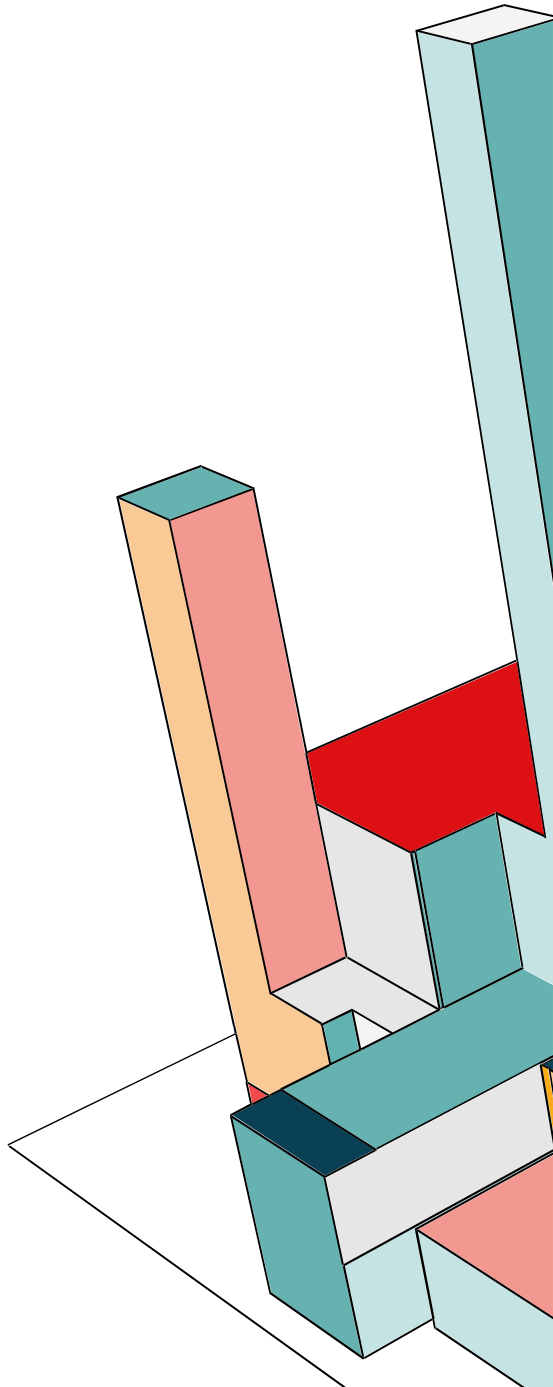




TASK

- Modify one of the models to obtain **higher-resolution output maps**
  - e.g. smaller patches, super-resolution decoder. **Ideas welcome!**





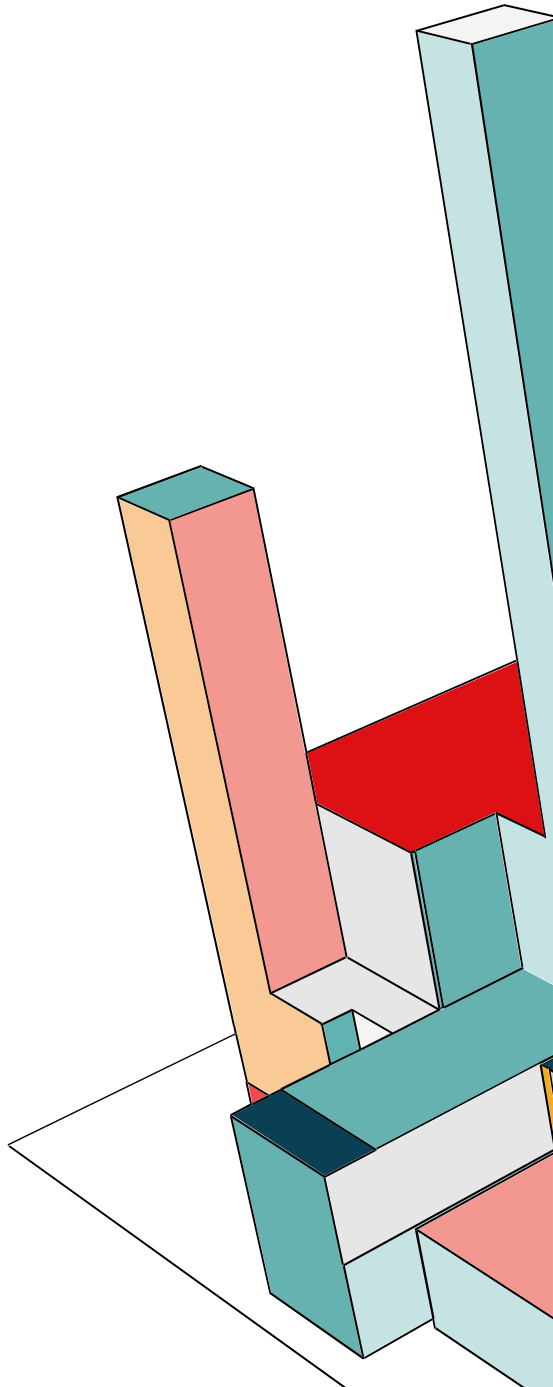
LIGHT MENTORING

Mentors

- Luca Barco *luca.barco@linksfoundation.com*

Weekly 30-min calls with students for the whole duration of the semester

Feel free to reach out via **Teams** or **email** at any time for any questions or doubts



POLICY

- Both project descriptions and implementations will be part of a repository group published on GitHub

- The repositories will be public

